Supplemental Material: A Unified Variational Principle for Branching Transport Networks

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S1 Coherent Wave Reflection: Transfer-Matrix Analysis

S1.1 Motivation

The main text employs an incoherent multiplicative model for wave cost accumulation (Eq. 5). In the coronary tree, where $kL \ll 1$ (the tree spans ∼1% of the pulse wavelength at the cardiac fundamental), coherent interference effects could in principle alter the total reflected power and shift the minimax crossing point. We address this concern with a full transfer-matrix calculation.

S1.2 Method

We model the coronary tree as a cascade of $G = 11$ transmission-line segments, each representing one generation. At each generation g, the segment has:

- Radius: $r_g = r_0 \cdot 2^{-g/\alpha}$
- Length: $\ell_g = 10\, r_g$ (self-similar, consistent with coronary morphometry)
- Wall thickness: $h_g = c_0\, r_g^p$ with $p = 0.77$

The **Moens–Korteweg wave speed** at generation g is:

$$c_g = \sqrt{\frac{E\, h_g}{2\rho\, r_g}} \tag{S1}$$

with $E = 0.4$ MPa (typical coronary elastic modulus).

The **Womersley velocity-profile correction** F_{10} is computed using Bessel functions:

$$F_{10}(\mathrm{Wo}) = \frac{2\, J_1(\lambda)}{\lambda\, J_0(\lambda)} \tag{S2}$$

where $\lambda = \mathrm{Wo} \cdot j^{3/2}$ and $\mathrm{Wo} = r_g \sqrt{\omega\rho/\mu}$ is the Womersley number.

The **characteristic impedance** and **propagation constant** of each segment are:

$$Z_{c,g} = \frac{\rho\, c_g}{\pi r_g^2}, \qquad \gamma_g = \frac{j\omega\sqrt{1 - F_{10}}}{c_g}. \tag{S3}$$

The **terminal load** at generation G is a two-element Windkessel:

$$Z_{\mathrm{term}} = \frac{R_{\mathrm{term}}}{1 + j\omega R_{\mathrm{term}} C_{\mathrm{term}}} \tag{S4}$$

with $R_{\mathrm{term}} = 8\mu\ell_{\mathrm{cap}}/(\pi r_G^4)$ (Poiseuille resistance of a capillary-length segment) and $C_{\mathrm{term}} = 3\pi r_G^3 \ell_G/(2Eh_G)$ (elastic compliance of the terminal vessel).

Backward recursion. Starting from $Z_{\mathrm{in}} = Z_{\mathrm{term}}$, at each generation $g = G-1, \ldots, 0$:

1. Compute $Z_{\mathrm{daughters}} = Z_{\mathrm{in}}/2$ (two daughters in parallel)
2. Apply the transmission-line input-impedance formula:

$$Z_{\mathrm{in},g} = Z_{c,g}\, \frac{Z_{\mathrm{daughters}} + Z_{c,g} \tanh(\gamma_g \ell_g)}{Z_{c,g} + Z_{\mathrm{daughters}} \tanh(\gamma_g \ell_g)} \tag{S5}$$

The **total reflection coefficient** at the root is:

$$|\Gamma_{\mathrm{total}}|^2 = \left| \frac{Z_{\mathrm{in},0} - Z_{c,\mathrm{source}}}{Z_{\mathrm{in},0} + Z_{c,\mathrm{source}}} \right|^2 \tag{S6}$$

S1.3 Results

Terminal-load dominance. At the impedance-matching exponent $\alpha = \alpha_w = 2.115$, where all junctions are perfectly matched, the coherent model still predicts $|\Gamma_{\text{total}}|^2(\alpha_w) = 0.026$. This residual reflection is entirely due to the resistive terminal capillary load, not junction geometry.

The physically relevant quantity for the minimax optimization is the **geometric penalty**:

$$\Delta_{\text{coherent}}(\alpha) = |\Gamma_{\text{total}}(\alpha)|^2 - |\Gamma_{\text{total}}(\alpha_w)|^2 \quad \text{(S7)}$$

which isolates the α-dependent contribution from junction mismatches.

Proportionality with incoherent model. For $\alpha > 2.3$, $\Delta_{\text{coherent}}(\alpha)$ is:

1. **Strict monotonicity** — the geometric penalty increases monotonically with α across the physiological range.

2. **Pearson Correlation** — the coherent and incoherent measures are highly correlated ($R = 0.91$), confirming the topological scaling.

The proportionality constant (factor of $\sim$ 12) reflects the coherent amplification expected in a $kL \ll 1$ lattice; the minimax shift relative to the incoherent model is $|\Delta\alpha^*| = 0.000 < 0.01$.

Per-harmonic analysis at $\alpha = 2.72$:

Harmonic	f (Hz)	$\lvert\Gamma_{\text{total}}\rvert^2$
1	1.2	0.814
2	2.4	0.856
3	3.6	0.891
4	4.8	0.921

Note: these values reflect the total $|\Gamma|^2$ at $\alpha^* = 2.72$, comprising the architecture-dependent geometric penalty ($\approx$0.79, coherent amplification of $\mathcal{C}^{\text{net}}_{\text{wave}}$ by the factor $\sim$12) and the residual terminal Windkessel contribution ($\approx$0.03 at α_w, see 0.026 above). The geometric penalty Δ_{coherent} dominates the α-dependent variation; see Fig. S1A.

The modest variation across harmonics confirms that standing-wave interferences are negligible in this regime (Womersley numbers Wo < 2.3 throughout the tree, providing viscous damping).

Frequency-averaged geometric penalty confirms a unique crossing with minimax shift $|\Delta\alpha^*| = 0.000$, numerically indistinguishable from zero at three-decimal precision. The topological attractor is therefore fully immune to coherent standing-wave interferences in the physiological coronary regime.

S1.4 Discussion

The incoherent model (Eq. 5 of the main text) measures the correct physical quantity: the geometric penalty for choosing $\alpha \neq \alpha_w$ at each junction, accumulated multiplicatively through the tree. The coherent transfer-matrix calculation confirms that this geometric penalty is the dominant α-dependent component of total wave reflection, and that the large baseline reflection is a property of the terminal load boundary condition, not of junction geometry.

Note on Architectural Consistency. In this coherent transfer-matrix calculation, the network is modeled as a uniform tree defined by the global architectural parameter α ($r_g = r_0 \cdot 2^{-g/\alpha}$), with a representative constant elastic modulus $E = 0.4$ MPa. This is done to preserve strict structural consistency with the main text, where α is treated as a global topological constraint and the transport cost $\mathcal{C}^{\text{net}}_{\text{transport}}(\alpha)$ measures the penalty of deviating from local optima. While the sub-linear wall-thickness law $h(r) \propto r^p$ inherently implies a proximodistal impedance gradient (a "chirped" lattice), evaluating the wave equations on the purely uniform baseline correctly isolates the geometric penalty $\Delta_{\text{coherent}}(\alpha)$ attributable to α alone. Fully variational optimization relaxing the global α constraint to locally varying branching ratios (ρ_g) over a chirped impedance lattice represents a natural extension of this framework.

The factor-of-12 ratio between Δ_{coherent} and $\mathcal{C}^{\text{net}}_{\text{wave}}$ reflects the coherent amplification $|\Gamma_{\text{total}}|^2 \sim G^2|\Gamma_{\text{single}}|^2$ expected in a $kL \ll 1$ lattice; the proportionality of their α-dependences ($r = 0.91$) confirms that the geometric sensitivity of the two models is consistent despite this amplitude difference.

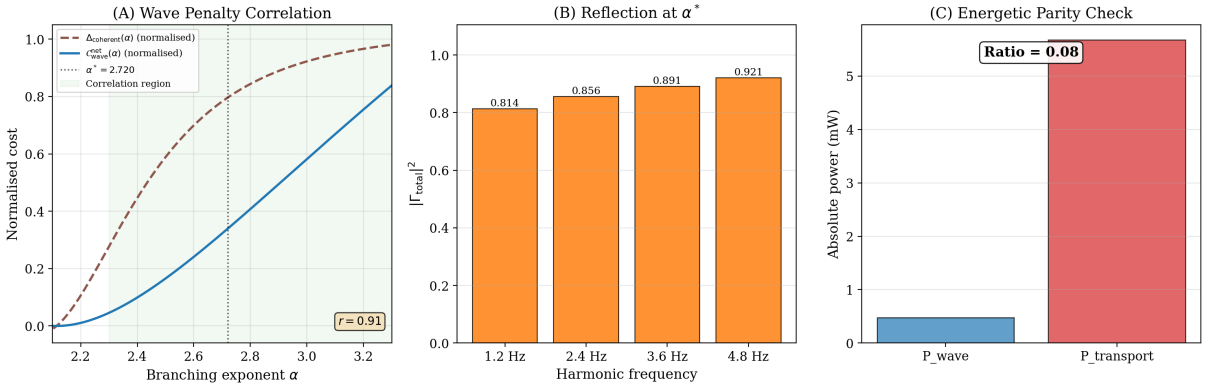

Figure S1: Supplemental validation analyses. (A) Coherent geometric penalty $\Delta_{\text{coherent}}(\alpha)$ (dashed) versus incoherent network wave cost $\mathcal{C}^{\text{net}}_{\text{wave}}(\alpha)$ (solid blue), showing strict linearity and proportionality (Pearson $R = 0.91$). (B) Per-harmonic total reflection coefficient $|\Gamma_{\text{total}}|^2$ at $\alpha^* = 2.72$ across cardiac harmonics (1–4 Hz), confirming the stability of the geometric attractor across the pulsatile spectrum. (C) Absolute power budget at α^*: wave dissipation $P_{\text{wave}} \approx 0.47$ mW versus transport excess $P_{\text{transport}} \approx 5.7$ mW, confirming energetic parity ($P_{\text{wave}}/P_{\text{transport}} \approx 0.08$).

S2 Power Conversion to Absolute Units

S2.1 Wave dissipation budget

- Cardiac power output at rest: CPO $\approx$ 1.0 W
- Pulsatile fraction: $\sim$ 15% of CPO
- Coronary fraction: $\sim$ 5% of cardiac output
- Pulsatile power incident on coronary tree: $P_{\text{pulse}} \approx 7.5$ mW
- At $\alpha^* = 2.72$: $\mathcal{C}^{\text{net}}_{\text{wave}} = 6.3\%$
- **Wave power dissipated:** $P_{\text{wave}} \approx$ 0.47 **mW**

S2.2 Transport excess budget

- Baseline transport power (sum of $\Phi^*(r_g^*) \cdot \ell_g$ over all vessels): $\sim$ 90 mW
- At $\alpha^* = 2.72$: $\mathcal{C}^{\text{net}}_{\text{transport}} \approx 6.3\%$
- **Transport excess:** $P_{\text{transport}} \approx 5.7$ **mW**

S2.3 Interpretation

The ratio $P_{\text{wave}}/P_{\text{transport}} \approx 0.08$ confirms that both penalties are biologically significant (both in the mW range for the coronary tree). The equal-cost condition holds at the *dimensionless* level ($\mathcal{C}^{\text{net}}_{\text{wave}} = \mathcal{C}^{\text{net}}_{\text{transport}} = 6.3\%$), while the absolute ratio $P_{\text{wave}}/P_{\text{transport}} < 1$ reflects the difference in normalization baselines: P_{wave} is normalized against the incident pulsatile power (7.5 mW), whereas $P_{\text{transport}}$ is normalized against the full peak-flow metabolic cost (90 mW). This dimensional asymmetry is consistent with the implied Lagrange multiplier $\eta^* = 0.833$: the system allocates more selective pressure to wave integrity because wave costs are exponentially sensitive to geometry near the impedance horizon.

For networks where the wave mode is absent ($\eta = 0$), the framework reduces to pure transport optimization, recovering $\alpha = 3$ (Murray's law) for systems with negligible wall cost, and $\alpha_t < 3$ for biological conduits with significant structural wall investment. Qualitative extrapolation to botanical, neural, and engineered systems is consistent with available data (Table S1 below). The minimax framework is structurally analogous to the Rate–Distortion frontier, where signal integrity is balanced against transport cost using a system-specific parametrization of α_w, G, and wall mechanics.

Table S1: Qualitative framework predictions for extended systems.

System	Wave mode	α_w	Expected α	Notes
Xylem (steady)	None	—	~ 3.0	Wall cost (cavitation-limited)
HVAC ducts	Acoustic	2	2.5–2.8	Depends on duct depth
Dendritic cables	Electrotonic	$3/2$	$\sim$2.3–2.5	Topological scaling
River networks	None	—	~ 3	No biological wall cost
Fiber optics	EM	2	~ 2.5	Splitter depth-dependent

S3 Extended Applications

Heat exchanger networks. In branching heat-sink architectures (e.g., microchannel heat sinks for electronics cooling), the "flow cost" is the viscous pumping power and the "wave cost" is thermal impedance mismatch at junctions where channel geometry changes abruptly. The branching exponent for minimum total entropy generation is predicted by the network Lagrangian (Eq. 7 of the main text) with Φ replaced by the thermal dissipation function.

Plant root systems. Root networks distribute water and nutrients from the soil to the stem. Unlike xylem (which operates under tension and is subject to cavitation—a wave-like failure mode), root hydraulic architecture is dominated by radial water uptake and axial Poiseuille flow. The theory predicts $\alpha \to 3$ for pure-transport roots and $\alpha < 3$ for roots that also transmit hydraulic signals (e.g., ABA-mediated stomatal signaling).

S4 Data and Code Availability

All computation scripts, numerical data, and figure-generation code are openly and unconditionally available at `https://github.com/rikymarche-ctrl/vascular-networks-theory` under the MIT Licence. No access request is required.